ActInf Textbook Group ~ Cohort 4 ~ Meeting 21 (Chapter 10 part 1)

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all right welcome everybody it's November 14th and we're in cohor 4 first discussion on chapter 10 so who wants to begin with some thought or question or any reflection about 10 any can raise their hand or just go for it but what they thought 10 was going to be or some component of what they remembered of it or just some quote that we could start with or question that they already wrote sure uh I mean just kind of at a cursory glance at this chapter it's just they're basically going through a variety of of other fields that may or may not be related to active inference and kind of comparing and contrasting uh I think it helps to understand what kind of um kind of trend or or or body of knowledge in which you know active inferences has its roots or where it's starting to assert itself as a as a distinct field um it's interesting learning a little bit more about for example uh the helm Holden view of like um unconscious inference as being like one of the key underpinnings of um active inference here um seeing more on the Basi and brain hypothesis and reinforcement learning and other fields that we see mentioned through the textbook but it's here that we get like a direct kind of comparison um and in addition to that I think one of the key overarching themes that they're trying to make is that active inference combines a variety of kind of traditionally separate or separated um cognitive functions like perception action planning attention and and so on that's the the title of the chapter is active inference is a unified theory um makes a lot of sense nice very nice so I um um there's something about the textbook overall um that um I found myself wondering about when I was reading chapter 10 um which is uh that there are many places in the book where it's not really clear if we're talking only about how the human brain functions versus how cognitive systems function and so there are places where um where it's almost uh it's almost it feels like a a tease like let's let's explore this is not just about the human brain this is about other cognitive systems but I think for people who uh the uninitiated and of course I'm one of those people at the moment but um it's it's tough right because I've created certain expectations about looking for how does this generalize to all kinds of cognitive systems but then in the last chapter the focus is and the wording is used for the rest of the chapter it's almost like we're going to introduce these Concepts like you would for a psychology textbook um and so they you know those various uh terms that are used in Psychology are addressed and how do we think about them from an active INF point of view then the section on social and cultural mechanisms I think needs to be either deleted or completely rethought um so when I kept going back to the figure in um chapter one I think it's figure 1.2 where it has a high road and a low road and in the center it has active inference and it has autop poesis and nich construction well Auto Poes and inch construction are brought up very infrequently throughout the textbook and I think there's an opportunity for chapter 10 to to go back to other places and each construction and and there are articles out there that I'm finding fascinating uh on the wording I'm still learning new words but what is this one called um joint agent I love this this article by ruinberg I don't know if this is a one that's um cited in in the book it might be uh but um joint agent environment systems and I think that that's the place where really cool stuff can be laid out uh for people who want to take this FR Market into the social sciences and anthropology thank you any other like thoughts on 10 or overall whether it's someone's like first time at 10 or repeating well I had I have another thought and kind I'm curious as to how others you know feel about this observation I don't know yet how where I would suggest to include this somewhere in the book but um if you know if most of the information that's presented in the book is backed up by data that's been gathered about the human brain then I think it would be interesting to point out that the human brain is actually a very new centralized nervous system in the history of living organisms and it's probably between 100,000 35,000 years old uh meaning that the act the present day uh the modern human brain organization presumably reached that point about that around that time 100,000 to 35,000 years ago so what I think is super interesting is that the brain the human brain that such a recent development has unveiled so much interesting information about um the ways in which we navigate the world and uh as sentient systems whatever that means right and said that that needs to be I think very clearly stated somewhere in the book what do we mean by a sentence system um but anyway I just thought I would bring that up and ask you uh other is like what do you think of that observation anyone can go or I'll give a thought Andrew go for it sure I I just I want to see if I understand the the question properly but I mean for me I think you know for example looking at the human brain and neurobiology looking at cortical micro circuitry for example finding a relationship between that and message passing the kind of things we find in like chapter five in the book I mean I think it's very I think it's very interesting I think it's it's fruitful and um I think that there have been empirical studies that are being done um and of course one of the biggest tricks is like to see like how like given we're external observers um we have to kind of see how the data behaves and figure out what is a reasonable proxy in the data for what's actually going on in the brain um so um I mean I I find a lot of validity in at least attempting to kind of make this connection between like the the brain however new it is um and and maybe not just the brain itself in terms of its circuitry but also like inferential processes trying to understand how those function in the brain or applying that as a kind of metaphor or analogy for looking at potenti uh entire Society although that's an area of that's the kind of work and active inference that I'm less familiar with so I don't want to overp speak on that um I I don't know if that was like a like a um you know onetoone like relevant response to your question so I'm just yeah I'm happy to keep talking about it if I understand you yeah I guess uh so I guess what I'm trying to uh appreciate like I don't really have anything super compelling to say other than to note or make the observation as context for this book I think is that the first centralized n uh nervous systems showed up in the evolutionary record I think about 570 million years ago and and so and the the book is about sentient cognitive you know s sentient systems presumably and so I had to do some digging and think well are we talking about uh centralized nervous systems that evolved emerged 570 million years ago and then you know from that perspective then I would uh I Marvel at the fact that you know humans have this incredible engine that evolved very recently and yet it has the capacity to not just do everything that's in you know I I've completely sold and believe in the whole active inference framework at this point you know I just think it's it's just revealing so much that I can know information about but I find it super interesting that this brain can actually externalize our generative models and that we can look at those generative models and that's why we we can do science right um and so it's not so much that I I think it makes the work less valid because we are such a recent development in human in evolutionary history but rather how this these processes phenomenally led to this brain that can do not only what you know being able to contemplate and think about our generative models but to actually create generative models through artificial intelligence and that is something about this book that's an Insight that can come out as one of a really important aspect of the book somewhere so that that's what I'm trying to convey thank you yeah these are great gray points here's a figure we often return to in this path integrals paper so it was like absent literature until it was drawn showing the continuity of the basil cognition and the different types things and so yeah there's that [Music] some fundamental aspects to that reflexivity there's a lot of it brings everything you know it it it opens up a lot which is why and and so I think this is a really important point that you're bringing up like do we approach the book is what it is 2022 textbook can never be changed they could release a different version we could generate a massive amount of educational material and develop ecosystem around this open source textbook Etc so so it's not about changing from how I see it what that book is but how it is approached and conveyed could run the gamut from an extension to a statistics course ranging to including deeper investigations like you brought up of the longer Arc of sentience and just allowing that to open more and into slower different regimes of attention yes in in this framework Daniel where does con context come into like in a in a formal way what does anyone think where does context come into play I'm S I want to make sure I understand the question again uh context in the form of do you mean as far as like active inference and and behavior goes or do you mean the field itself like it's context um okay um I mean I I'm let's go back to the 1.2 where you know it's the high road and the low road where in this in this world doeses context come into this science there's probably a ton of answers but one short narrow answer is self-organization or just broadly the partitioning or differentiation that allows us to say self versus nonself or just any kind of continuation that's the context that's the niche so the first particular partition that enables us from not being able to say anything about that thing that thing not existing to it being a thing it comes into description um as contextualized another related point is this as if instrumentalist view that the the it doesn't take on um needing to provide an account of how whatever context is applies to the rat in the maze so that's why it centers potentially that self-reflexivity of the Observer because the model not to say it's boring not to say it's simple but it's just an artifact resulting from a different process yes um so so I okay but when you know when you're talking about something like how do I want to ask this question um so you use the word artifact I'm just going to stick for now with the idea that that artifact in 1.2 is applicable across all sorts of contexts across all kinds of agents and I think what what I would like to know is where where is the I know I shouldn't use that word but where's the boundary I was going to say Mark of blanket I'm not going to use it but where's the boundary um in other words in what systems does this arti fact not is is it not applicable yeah very common question about the applicability or the scope of this type of modeling and and so if it's left us as as an open question that's okay and I won't you know push any further but I it's really been does anyone want to give a thought yeah I I can give a thought and I'm uh in cohort five so I'm just uh I guess piggybacking off this um but um I think that one of the you know active inference is a model of understanding really any natural system that you want to look at um that behaves in under the certain Dynamics um it could be applied to just about anything so the context is really going to be up to the person using it so you could have any context and you know try to draw the boundaries anywhere and um how you draw those boundaries is going to determine what the organism or particle that you're looking at is and what the environment is so I I think that's one of the cool things about active inference is it's so broadly applicable and where you draw those boundaries makes all the difference in how the models turn out thank you that was really help one note on this is figure 1.2 is not a formal argument it's more evocative so these different terms might apply in different ways to to different systems like planning as inference not all systems plan and then even again pulling back to the instrumentalist layer you don't have to model the system as planning but maybe you do model even an inment system as doing some kind of extended planning or vice versa so they're not requisite they're they're just uh some train stops on the low road and the high road and I think that is a layer that that does kind of make sense to pull back to not that this is exhaustive or um a Panacea but it's just a framing that helps organize the the book and the work and also situates active inference at the intersection between actually building something from the ground like what dennet might call the cranes approach and the tethering or the association to the high road principle the Skyhook in that same kind of denit metaphor so that there's something that can be realized with real software tools and real relationships between the math and the Theory and the implementations finding new useful through and throughs and therefore having nothing to say about any given Thing by virtue of being applicable to any modeling situation Brahman uh yes Daniel I was just um the figure 1. two I I I view it as you as you said sort of a you not not a fixed thing not really a a a pathway thing but but in ways a lot of those things were historical developments in the way people scientists and philosophers were considering um the progression of understanding of the sentient being or the we use human being in its environmental Niche so that sort of constant question of how is it that we persist how is it that we survive what is it we do how do we adapt and I keep going back and I was talking to you before about 10.3 and Daniel dennett's comment in 1978 about the necessity to to Really model the whole of Guana which is really about how do we understand the human being as a as a whole and so there's a lot of I mean and I also there's a distinction I think between this conversation on modeling something and a conversation on um understanding the Adaptive Dynamic nature of an evolving creature in its environmental Niche so in a way I sort of separate out those things was a modeling sort of the acting on the consideration of how it is that we continue to survive or not survive whatever it might be so I mean one thing I think that I've learned in my travels through active inference is its connection historically to a lot of thought from a lot of people and a lot of um consideration of of what we do how we do it and that's including our our our sort of evolving ability to be cognizant of ourselves and understand how we are actually making decisions so it's that stepping back out of our precisions actually um enabling us to fine-tune our own basic mechanics so that we can actually see ourselves clearer in our environments so that we can then function to survive and make better choices and then we can begin to sort of model that I suppose but I always have a question I'm rambling on a bit but I also I always have a question about someone creating a model when they actually have a biased view of which is which is the nature of it a biased view of what they're what they're doing so I think it's a um the distinction needs to be made between the actual uh what the fre fre energy principle states which is we must reduce or minimize uh free energy and the actual process theory of active inference in which we we bring that into um pragmatic play in our lives as a whole being in our Niche and then wrapped in that is then I think we can then subsume all of those other questions that arise whatever I just said wow but no but anyway one point two is that I mean if you go into each of those different areas autop poesis um predictive coding blah blah blah you know they they've all got a sort of a progressive historical evolutionary type footing um on which um I think active inference rests and um I even found another person in America that I was going to speak to about alre as I don't know if you've ever heard of him but um yeah there there's like a you know it's as if as a as a whole being we're we're evolving we're evolving to be able to recognize ourselves and uh what we do in our Niche I'll stop there yeah that's a very interesting point evidence by the perspectives people bring the skills that are required the kind of collaborations that would entail skillful usage a lot of different aspects to it yeah yeah and that's sort of like one of the follow-ups after the well the scope question is it applicable to Any Given system and it's like well the question presupposes if it's a system that's that someone's thinking of then yeah that's the system just go ahead and describe it so people try to hold that out like as if it's in a rug pull to talk well it could apply to everything it's like yeah um and then from there the natural question is like well then what does it do and the whole iguana it's funny because people make what they believe are Justified true belief assertions about human behavior and this molecule does this and it's like not not to say that smaller brains are simpler of course I'm always bringing up the insect brains here but think about how many experiments go into determining the role of like neurotransmitter at a certain synapse and so it's like way less evidence in humans obviously important everyone cares but then to overstate the

Precision we're not even at the whole iguana this is the whole chapter 7 tze it's like that's that is whole in what it is it's a pedagogical example and then by iterating with a real animal in the Maze then there's more and more and and also becomes clearer and clearer how the map's not the teritory from the first draft to to the one millionth draft map's not the territory under whatever constraints the map and the territory co-evolve under so it's are the the questions that that help make chapter 10 actually about connecting the dots and about anything more than just a literature review on psychological and philosophical topics and actually so when you you brought up the the word bias right now would it be a correct interpretation of generative models to say that um I think active inflence what it's one of the implications that that seems to make sense to me is that that we are constantly in the process of creating biases uh because by creating biases we um you could say that a bias is just a model in our heads of the world and in science we're constantly coming up with these models and but then we get a chance to discuss them and put them at risk um but that but that's just kind of an example of how powerful I think the power of this perspective that it actually allows me to think about something like bias that we talk about all the time and not just well it's just bias well what is bias I'd like to read a little section from a short piece that I wrote so basically um active inference Learners often look at representations such as figure 4.3 from the 2022 textbook or figure 1.2 or figure any of the textbook seemingly expecting to find sophisticated phenomena such as affect or narrative reflexivity in or on that representation two key aspects of my response in that situation first if active inference were to include even relatively simple cognitive phenomena learning or tension in its essential particular formulation particular partition this would explicitly restrict the scope of analysis to systems with that character and silently diminish our Capac to distinguish across that difference second natural language or conceptual descriptors of cognitive phenomena are secondary attributions better understood as relational rhetorical associations assertions using a pattern language of phenomena about a given realized generative model rather than fundamental or intrinsic aspects of the system of Interest itself not to be verbatim but I think it's a really important Point like bias has a narot technical definition you'll see in a statistics textbook it's what biases the data and does anything different than a pass through it's just a feature of how discriminatory a signaling algorithm is and then of course there's broader senses and and alignments with with with broader um usage of those terms just like with so many of the terms in the actin ontology and I think it's relevant that especially when talking about the broader systems of Interest or the phenomena that there there are not the basil low road they're not the assertion they're not what actually implements the system they're descriptors of systems with observers which is the the setting where more can be done and and it's a better discussion setting when it's assertion based not like not not to go too hard on this one point but I think that that that puts the space as us as generative model interpreters in between us and any created artifact without that articulation we're we're just not set up to have a productive epistemic conversation thought on that and I like that quote that you showed Daniel about the comment of the vagueness and difficulty with philosophical questions is intentional so that it's a generalizable model we don't want to be pigeon rolling it um but I I think bias and this is my perspective on it is a part of the learning process and if you're learning anything that you hope to be generalizable then when it's accurate it was useful information and when it's in a situation where it's not accurate then call it bias and anytime you're trying to generalize something you are creating a bias that you hope is going to be helpful and useful at some future point But as soon as it gets out of that context where it's helpful and useful we just call it bias it's skewing your understanding in a certain way yeah that's very important like if someone says that was a biased decision to think like how could they violate this or that but if it was like no they picked up on subtle cues it was a biased decision ision be like wow great for them in that situation so what what is really the discussion not even to that topic but just this is that that's stepping back from even our words and our symbols how does that really play out were Learners on that all all with each other in the textbook group there is a technical part to active but I think conversations like this chapters like this they show it's not simply that way the whole low road could be ripped out and replaced potentially even in the the message passing live stream right now with Magnus kudal um like generalized free energy generalizes variational free energy and expected free energy so those are some of the main topics we talk about about like equation 2.5 and 2.6 in this textbook but if those formalisms last for several years to to decade then they'll have been long lived even by things not accelerating standards so there's just a lot of interesting pieces like as the textbook recedes then do do we continue to drive back to the textbook do people want spaces of more open straightforward I don't know something in between all of the above of course um personally I need more time to to think about that I I'm still there's so much in the book that I still I'm processing and learning like that's kind of the funny joke at the end of the book making the the extracurricular claim that active inform is not something that can be learned purely in theory that's why it's such a great I I love that about it because honestly when when I as I read I have to pause and and then step away and think about how to S supply to this system that system other stuff I'm thinking about and then I come back um but you know

I'll just share with you guys uh that um so I've been studying Buddhist psychology for decades and what one of the things that is really mindblowing to me is how the very high level Buddhist psychologist um scriptures um are so much have very much to do with active inference yeah so so yeah I mean I I guess I'm just thinking about applications and then going back to the ideas digging deep and trying to master very complex Concepts in this second way Rin um that that's can I can I just ask a question um did you just say with the Buddhist psychology that um that active inference was congruent seems to make sense of of a lot of the philosophy was that what I heard well I'll I'll give you a spe specific example um so for ex so in the um so Buddhist psychology is very also very applied it's uh very abstract and very applied and so one of the applications is meditation so what is meditation what you do with your mind when you're meditating and essentially what you're doing is is uh putting at risk all your generative mod that the way that you find the idea behind that you know it's it's called um investigating your mind it's not you know if you have a horrible thought about killing somebody you know it's not that you just say oh no stop that thought it's examining why do I have that thought and then you you ask yourself that but to me when I I was reading about active inference is about yeah we create this generative models like since I'm an an apologist and I worked on these these different cultures I mean I I've lived in cultures where people will say we men can kill women and not be punished okay well that's a model that these people carry in their minds right and so in Buddhist psychology it's what is the the default kind of tape layer that's going on in your mind that's driving everything that you do that you never think about and that's what it is that's it's a simp it's very simple and uh it's very consistent with active inance glad you think so yeah there's like several articles and lines of research with like attention meditation Buddhist psychology different knowledge Traditions so that's definitely very interesting bronwin um yeah well because the reason I ask is is that that I am an Alexander technique teacher which most people wouldn't know what that was but the Alexander technique has a a history but it goes back in time to so 100 years ago with uh John dwey who was an educationalist philosopher in America and um I think it was stated once on on this book club that you know Jew Jewish language now uh is is quite relevant and and particularly if you read it from an active inference perspective and that's what I have been doing and and as well with the Alexander technique and it doing that having that view or that so it's maybe it's a preference that that seem things seem to have much more meaning and uh are much clearer to to be able to understand um that was just one point the other thing with about biases was I was just wondering is um is that that sort of discussion about bias within

within the um the mechan the basic mechanics uh we we have preferences and preferences are what keep you where you are and for me preferences are biases within the Precision mechanics pretty much in there and because it's an unconscious inference is is pretty much unconscious unless we um do something like meditate uh or pay attention uh investigate then it remains unconscious but but we probably have the ability to cognitively um delve a certain degree there's a certain penetrability to be able to that's what Jack of how talks about penetrability to be able to actually investigate a little further into our unconscious inferences um to be able to begin to understand why we why we think something but then a step on from there is that a generative model is made up of an action perception cycle and I don't it's just a comment I could be could be my preference that um a lot of discussion tends to um in in book club tends to and even within the broader literature uh remain in a sort of a passive State um in a perceptive state without the recognition that there's an inseparability of action and perception uh which I think is what's at the basis of active inference we we cannot separate the two so that with with anything we think or do it is an a perception and an action and um and I think it's much much more expansive than people's conceptualizations of how we actually function in the world a lot of our actions are UNC unconscious when their habits they're habitual when really not present to to what we're doing meditation and mindfulness meditation Buddhist Meditation is is one method of of beginning to investigate that act that action perception Loop that that that enables us to to function at every minute of the day really as as well as what I consider the the Alexander technique uh the Alexander technique in its true form uh brings action and perception and thinking together so that it's based on psychophysical um re-education process but I I just wanted to put forward I suppose that that the whole body the whole body seems to get broken up a bit in these discussions and and the action perception cycle cycle is that it's an intwined uh Inseparable functioning and I think the ability for for us to really cognate that to really sort of understand that is what what active inference uh hopefully will and I think is coming bringing to light it's a thing that draws people to it but preferences not biases yeah one interesting point there to seal that point is internal and external is just pick aside but sense and action are identically considered by the across any boundaries like and also I find it interesting that the Mew for internal state it is almost like if you rotate this like clockwise or counterclockwise 180 degrees that it's like it it looks like the symbol on the other side to to just show that 's kind of like a mirroring but but not a an equivalence but that's the interface that that enables there to be that Andrew sure I just wanted to speak on the I'm a practicing U Buddhist and and meditate regularly and have had a lot of thoughts about this although I haven't become too strongly haven't gone too deep into the literature on active inference and meditation but I I very much agree with this idea of um meditation is a kind of like practicing one's attention um paying attention to one's thoughts whatever is going on internally a lot of forms of meditation involved reading the body and so practicing interoception and um and with that there's a kind of I I don't know uh almost like a an attempt at overcoming something like habit like I'm getting back the models like you have your e uh Matrix versus G and I think by practicing by being able to have that sort of meta awareness I suppose around what your thoughts are what your desires are in in Buddhist terms or your preferences as far as the C Matrix goes and trying to figure out like what what are those about what am I going for do you have certain things that you want that you know you shouldn't want and like trying to overcome those um yeah I I think a lot of it involves like practicing one's attention that is I guess Precision optimization like trying to certain um yes and certain signals that you're receiving and and instead switching your attention to to others that might actually help you realize your your new preferences um as opposed to maybe whatever like you to the the negative habit before something and um but with Buddhism you have this notion of like a desire being kind of endless right um and so like it can you know kind of take you over if uh in the extreme and I'm not sure where that lies as far as um you know trying to use um P mdp to model that as a process or something goes you know but uh yeah it's all it's all very very interesting so I kind of just wanted to chime in on on that bit Yeah Yeah like just to connect the the the references a little bit in the textbook here was that kind of figure you alluded to 54 with the Seesaw or the Continuum between the total pass through of habit defined however and then the total sharpening to an arbitrary function which function that's life SL modeling but the this arbitrarily defined one such that any sharpening we can fit within this framework it doesn't even have to be one that is created by the modeler that's why it's a descriptive view from the outside and then here in the saned Smith uh 21 paper live from 28 this kind of demonstrated the plausibility in the scheme for nested awarenesses because just having attention or bias filter of one depth is just like sharpens out let's just say edges from a static image so successive layers of bias signal the noise could could pull out edges or or hallucinate them um but here it kind of pointed towards the ability that there could be awareness of awareness and and this sort of more nuanced view on phenomenology which could be understood just could be Whimsical could be different culture it could be different individuals just to show that it could be created in terms of this broader topic of regimes of attention attention as internal action internal action as covert action so using the same apparatus that's used for the overt action and like the planning where to walk turning that to apply to potentially sequences of attentions a je uh yeah so um I had a different question not related to sort of meditation so it's a little different but I'll go for it okay okay uh I was just wondering like how exactly is sort of uh I guess like Free Will and measurement Independence sort of how's that related to like the free energy principle and active inference like are like do behaviors follow like certain attractor States at certain times and does that mean that like when you're taking a measurement of something uh the choice between that measurement is actually like the choice between that measurement is statistically dependent uh with sort um like you kind of like uh if you're going to pick you know between A and B was that choice sort of determined if sort of you know our decisions themselves follow sort of this principle of least action with like the free energy principle and all of that so yeah I was just wonder like how is free will ort of formalized in uh active inference big question just so that a few other people can ask here's the shortest answer look it up in this Koda there's been a little bit of discussion on it people have come down hard on one side or the other people have put their professional careers on the line one side or the other science shows this science shows that that's how I see that debate largely at this point so active inference doesn't actually like tell us it doesn't lean towards one side as of yet or it doesn't have a system that it's based upon other than historically but it doesn't have a system indicated so it's it's expressionless about topics like the relevance of a given thing in a given situation for a given thing it's multiple layers of productivity by the modeler and the the contextualize before that's even a complete sentence so I think it's just carton horse or something like that but but understandable and topically Associated and historically Associated um but we we should continue to talk read theoda and write or add more and obviously it's an ongoing discussion so I'm not saying it's permanently one way or the other I'm just saying that you will find Discord messages and Kota and transcripts of people vehemently believing both things within the last two months okay okay yeah um everyone who wants to how about just a closing round of thoughts on this like first discussion on 10 so bronwin then um Magdalena then anyone else um just very quickly um in terms of active influence going forward you you discussed it a little bit before in terms of what you know sort of is the booking 22 what happens in in the future um I just wonder it seems to me that from very different areas in life people get their teeth they they hear about active inference and they get their teeth into it and it's really hard to Let It Go like it's just like a dog with a bone and um and I can't see how it it can do anything other than than evolve and become um not not denser but sparer in a way um a much more simple um Theory to be able to understand H sent in Behavior I just wonder what people think about that and what their their view on the um the future of active inference is I mean for me it's something I I haven't been been able to let go of and it just raises more questions day after day so yeah I'll just leave it there yeah everyone can can give a closing thought on that or whatever they want um yeah my closing thought is um that uh I first want to learn active entrance really well I want to understand it fully I think it has a lot of promise and I I every day I do a little bit of translation into the um scientific question C I'm working on um but I feel the same way you do I just feel like this this is just the beginning of a very long journey it's gonna take a lot of work sorry about that any any other closing thoughts on 10 or um yeah uh I think active inference is rather huge in this uh in the context of chapter 10 presenting the theory as a broad unified theory I think it's going to go in many directions for whoever wants to start modeling in that direction I myself discovered active INF like just a kind of hobbyist interest in psychoanalysis and then I found uh the work of Mark SS in neuros psychoanalysis and then one thing led to another and here I am wanting to read Carl friston's work uh instead almost um and and seeing how this is being applied in feels as diverse as Psychiatry to robotics so um yeah I think it really I'm more interested in seeing what kind of empirical work like what kind of direct applications um people carry out um with this Theory and I think that'll probably end up shaping the field even more closing but yeah awesome asil do you want to add anything yeah today we speak about Daniel denet and Buddhism I think one thing is common between Daniel D and Buddhism both of them deny self uh in the meditation um almost you you will lost self and Daniel Den introdu s is like a a school um for example after 20 years in the school everything will be changed like teacher student bench everything just they have same name and this General model uh is I think exactly whole IG this don't look um don't suppose self this active inference make General model for artificial intelligence and people braing and everything generative not uh um with some um um some abstract idea I think so thank you I think that's a super important point about how it's purely synthetic and that is being explored more recently so however satisfied and excited we were previously it continues to develop and as with all of our discussions this one probably opens up more questions than not but these were all really great points chapter 10's kind of Epic but it's also like symbolic and it is where we jump out you know we're going to close this meeting we close that book and then you do something next so that's really where like breaks the fourth wall breaks the fifth wall all these like interesting Dynamics of our engagement with it and all the fun things we get to do in textbook group yeah though we'll have another discussion next week earlier then this time in two weeks we will talk about projects or future textbook groups right now already form yeah is accurate so you can fill out the form you can give any other information but other than that we're just kind of chilling cruising down the end of the year last few scheduled live streams have a quiet December then come back in 24 anyone's welcome to be a facilitator for textbook group or we can explore like other things so thank you see you next time bye thank you Daniel